Exercises — Semi-logarithmic graphs and extrapolation

3A — 30/10/2026

Newton's law of cooling: $T - T_{room} = (T_0 - T_{room})\, e^{-kt}$, *so* $\ln(T - T_{room}) = \ln(T_0 - T_{room}) - k\,t$.

Exercise 1 — Which graph is a straight line?

A potato cools in a room at $T_{\text{room}} = 21\,^{\circ}\text{C}$. Its temperature is measured every 5 minutes.

t / min	0	5	10	15	20
T / °C	61.0	49.3	41.0	35.1	31.0

(a) Complete a table of $T - T_{\text{room}}$ and $\ln(T - T_{\text{room}})$ (3 decimal places).

(b) Is the graph of T against t a straight line? Is the graph of $\ln(T - T_{\text{room}})$ against t a straight line? Explain using the values.

(c) Find the gradient of the second graph and the value of k in min^{-1}.

Exercise 2 — Gradient of the best-fit line

A student plots $y = \ln(T - T_{\text{room}})$ against t (in s) for a cooling potato and draws the best-fit line. Two points read *on the line*, far apart, are (200 s, 3.05) and (1000 s, 2.65). $T_{\text{room}} = 20\,^{\circ}\text{C}$.

(a) Calculate the gradient and k, with units.

(b) Calculate the intercept at $t = 0$ and the temperature T_0 of the potato at Start.

(c) Calculate the time $1/k$ and the half-life of the difference $T - T_{\text{room}}$.

(d) Why must the points of the gradient triangle be taken *on the line*, far apart, and not from the table of data?

Exercise 3 — Extending the line back

A potato leaves a water bath at $T_{\text{bath}} = 45\,^{\circ}\text{C}$ and is left in a room at $T_{\text{room}} = 22\,^{\circ}\text{C}$. The best-fit line of $y = \ln(T - T_{\text{room}})$ against t is

$$y = 2.90 - 3.0 \times 10^{-4}\, t \qquad (t \text{ in s},\ t = 0 \text{ at Start}).$$

(a) Calculate $\ln(T_{\text{bath}} - T_{\text{room}})$.

(b) Find the time at which the line reaches this value. How many minutes before Start did the potato leave the bath?

(c) Check that your answer is sensible: calculate the temperature of the potato at Start. Is it below T_{bath}?

(d) Would you expect the real time to be longer or shorter than your estimate? (Think of how the centre of the potato cools at the beginning.)

Exercise 4 — Two traps

(a) A student uses the data of Exercise 1 but plots $\ln T$ against t instead of $\ln(T - T_{\text{room}})$. Calculate the gradient between $t = 0$ and $t = 20$ min and compare it with the value of Exercise 1. What error would this make in k and in the half-life?

(b) The Arduino thermometer reads $+1.0\,^{\circ}\text{C}$ too high everywhere (zero error). The student reads both T and T_{room} with it. $T_{\text{bath}} = 50\,^{\circ}\text{C}$ is measured with a correct reference thermometer; the true room temperature is $22\,^{\circ}\text{C}$, the gradient is $-3.0 \times 10^{-4}\ \text{s}^{-1}$. Does the zero error change the gradient? And the intercept? How does it change $\ln(T_{\text{bath}} - T_{\text{room}})$ and the estimated time?